A Ray Obstruction to Global C_b Approximation for Rank-One Sigmoid Frechet Networks

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Status

This packet records a candidate counterexample, likely valid. It disproves a natural global supremum-norm strengthening of the rank-one sigmoid activation example in Benth–Detering–Galimberti, arXiv:2109.13512v4. It does not contradict their compact-open density theorem.

Source Passage

Remark 2.2 of the source paper says that it would be tempting to work directly in $C_b(\mathfrak{X};\mathbb{F})$, endowed with the supremum norm, after imposing boundedness assumptions on the nonlinearity σ.

> **Remark 2.2.** *It would be tempting, albeit more challenging, to establish our universal approximation result (Thm 2.3) in a "global" setting, namely to work directly in the space of bounded continuous functions $C_b(\mathfrak{X};\mathbb{F})$, endowed with the supremum norm (upon imposing suitable boundedness conditions on the non-linearity σ), rather than staying at a "local" level as we are doing now.*
>
> *The main obstruction that prevented us from employing this approach is explained by the succeeding observation: If we aim at following Cybenko's blueprint [15] (refer to the proof of Thm. 2.3 below) to establish our result, then in that case we would be required to work with the space*
>
> $$rba(\mathfrak{X}) := \{\mu : \mathcal{B}(\mathfrak{X}) \to \mathbb{F};\ \mu(\emptyset) = 0, \textit{finitely additive, finite and regular}\}$$
>
> *which is known to be the dual of $C_b(\mathfrak{X};\mathbb{F})$, i.e. $C_b(\mathfrak{X};\mathbb{F})' = rba(\mathfrak{X})$. Dealing with finitely additive measures is more involved, because many standard results from classical measure theory cease to hold. In particular, at this stage it is not clear to us to envisage a suitable set of conditions that the non-linearity σ must satisfy in order to be discriminatory (see Def. 2.6).*
>
> *Nonetheless, we deem this potential extension of our result to be interesting and worthy to be explored (most likely by deviating completely from Cybenko's strategy of proof), and we hope to be able to come back to this question in the future.*

The activation class treated here is the source paper's one-coordinate sigmoid example: after choosing $\psi \in \mathfrak{X}' \setminus \{0\}$, $z \in \mathfrak{X}$, and $\beta \in C(\mathbb{R};\mathbb{R})$ with

$$\lim_{\xi\to\infty} \beta(\xi) = 1, \qquad \lim_{\xi\to-\infty} \beta(\xi) = 0, \qquad \beta(0) = 0,$$

the source notes that

$$\sigma(x) = \beta(\psi(x))z$$

still gives compact-open density in $C(\mathfrak{X};\mathbb{R})$.

consider a function $\beta \in C(\mathbb{R};\mathbb{R})$ such that

$$\lim_{\xi\to\infty} \beta(\xi) = 1, \ \lim_{\xi\to-\infty} \beta(\xi) = 0, \ \beta(0) = 0,$$

and arbitrary $z \in \mathfrak{X}$ in order to define

$$\sigma(x) = \beta(\psi(x))z = \beta(x_1)z, \quad x \in \mathfrak{X}$$

which still enables us to conclude that $\mathfrak{N}(\sigma)$ is dense in $C(\mathfrak{X};\mathbb{R})$. Example 4.4 below extends this example for more general choices of ψ. A natural question now would be to find "optimal" β and z such that the convergence of the approximation to the function we want to learn is "fast".

Example 2.13. *The above example can be extended to an activation function that operates on infinitely many different directions $z_j \in \mathfrak{X}$. More precisely, let now $\mathfrak{X}$ be a real Banach space with norm denoted by $\|\cdot\|$. As above, we consider an arbitrary $\psi \in \mathfrak{X}' \setminus \{0\}$. Moreover, suppose we have a sequence $(\beta_j)_{j\in\mathbb{N}} \subset C(\mathbb{R};\mathbb{R})$ such that*

$$\lim_{\xi\to\infty} \beta_j(\xi) = 1, \ \lim_{\xi\to-\infty} \beta_j(\xi) = 0, \ \beta_j(0) = 0, \ j \in \mathbb{N}$$

Transcription of the Target

The relevant part of Remark 2.2 reads, in substance: it would be interesting to establish the universal approximation result in the global space of bounded continuous functions $C_b(\mathfrak{X};\mathbb{F})$, with the supremum norm, under suitable boundedness assumptions on σ. The authors identify the dual space $rba(\mathfrak{X})$ and finitely additive measures as the obstruction to a direct Cybenko-type proof.

The Counterexample

Theorem 1. *Let $\mathfrak{X}$ be a nonzero real Frechet space. Let $\psi \in \mathfrak{X}' \setminus \{0\}$, let $z \in \mathfrak{X} \setminus \{0\}$, and let $\beta \in C_b(\mathbb{R})$ have finite one-sided limits*

$$\beta(+\infty) := \lim_{s\to\infty} \beta(s), \qquad \beta(-\infty) := \lim_{s\to-\infty} \beta(s).$$

Define

$$\sigma(x) = \beta(\psi(x))z, \qquad x \in \mathfrak{X},$$

and let

$$\mathfrak{N}(\sigma) = \operatorname{span}\{x \mapsto \ell(\sigma(Ax+b)) : \ell \in \mathfrak{X}', \ A \in \mathcal{L}(\mathfrak{X}), \ b \in \mathfrak{X}\}.$$

Then $\mathfrak{N}(\sigma)$ is not dense in $C_b(\mathfrak{X};\mathbb{R})$ with the supremum norm. Indeed, there is $f \in C_b(\mathfrak{X};\mathbb{R})$ such that

$$\inf_{g\in\mathfrak{N}(\sigma)} \|f-g\|_\infty \geq 1.$$

Idea of the Proof

The source's one-coordinate sigmoid activation collapses every neuron to a bounded scalar sigmoid of one affine functional. Along any fixed ray $t \mapsto tv$, every such scalar input is affine in t, so the finite limits of β force every neuron, and hence every finite network, to have a finite ray limit. But $C_b(\mathfrak{X})$ contains bounded continuous functions that oscillate forever along the same ray, for example $x \mapsto \sin(\varphi(x)^2)$. Such a function cannot be uniformly approximated by functions with a ray limit.

Proof

Proof. Because $\mathfrak{X}$ is a nonzero Hausdorff locally convex space, its continuous dual separates points. Choose $v \in \mathfrak{X}$ and $\varphi \in \mathfrak{X}'$ such that $\varphi(v) = 1$. Set

$$f(x) = \sin(\varphi(x)^2), \qquad x \in \mathfrak{X}.$$

Then $f \in C_b(\mathfrak{X}; \mathbb{R})$.

Let $h(x) = \ell(\sigma(Ax + b))$ be one neuron. Along the ray $t \mapsto tv$,

$$h(tv) = \ell(z)\,\beta(\psi(A(tv) + b)) = \ell(z)\,\beta(t\,\psi(Av) + \psi(b)).$$

If $\psi(Av) > 0$, this tends to $\ell(z)\beta(+\infty)$. If $\psi(Av) < 0$, it tends to $\ell(z)\beta(-\infty)$. If $\psi(Av) = 0$, it is the constant $\ell(z)\beta(\psi(b))$. Thus every neuron has a finite limit along tv as $t \to \infty$, and therefore every finite linear combination $g \in \mathfrak{N}(\sigma)$ has a finite ray limit

$$L_g := \lim_{t \to \infty} g(tv).$$

On the other hand,

$$f(tv) = \sin(\varphi(tv)^2) = \sin(t^2).$$

Let

$$t_n = \sqrt{\frac{\pi}{2} + 2\pi n}, \qquad s_n = \sqrt{\frac{3\pi}{2} + 2\pi n}.$$

Then $t_n, s_n \to \infty$, while

$$f(t_n v) = 1, \qquad f(s_n v) = -1.$$

Suppose, for contradiction, that some $g \in \mathfrak{N}(\sigma)$ satisfies $\|f - g\|_\infty < \varepsilon$ for some $\varepsilon < 1$. Passing to the limits along the two sequences gives

$$|1 - L_g| \le \varepsilon, \qquad |-1 - L_g| \le \varepsilon.$$

Hence

$$2 = |1 - (-1)| \le |1 - L_g| + |-1 - L_g| \le 2\varepsilon < 2,$$

a contradiction. Therefore no $g \in \mathfrak{N}(\sigma)$ can approximate f within supremum error < 1, and the distance from f to $\mathfrak{N}(\sigma)$ is at least 1. □

Verification Notes

The proof is purely analytic and uses no computation. The only topological vector space fact used outside the source definitions is the standard separation property of nonzero Hausdorff locally convex spaces: for a nonzero point one can find a continuous linear functional not vanishing at that point. This is a direct Hahn–Banach consequence.

The finite limits of β are exactly the needed input. Monotonicity, sigmoid shape, and the normalization $\beta(0) = 0$ play no role in the negative global result.

Novelty and Literature Check

The run indexes were searched for arXiv:2109.13512, the source title, “Frechet spaces”, “Fréchet spaces”, “neural”, “C_b”, “supremum norm”, “finitely additive”, “discriminatory”, and the formula “sigma(x)=beta(psi(x))z”. No exact previous packet was found. One adjacent run entry concerns arXiv:2105.02095 and a different Banach-valued neural network question.

External web searches on 2026-06-20 used the source title together with “C_b”, “supremum norm”, “finitely additive”, and “discriminatory”, plus targeted searches for “sigma(x)=beta(psi(x)) z”, “sigmoidal C_b not dense”, and “sin(xˆ2)” ray-obstruction language. These searches returned the source arXiv page but no separate answer or matching counterexample. Novelty confidence is moderate: the argument is short and may be folklore in the classical $C_b(\mathbb{R}^n)$ setting, but no source was found that records this obstruction for the Frechet-space networks of arXiv:2109.13512.

Limitations

This is not a classification of all bounded nonlinearities σ that might work in global $C_b(\mathfrak{X})$. It shows that the source paper’s concrete rank-one bounded sigmoid activation, despite giving compact-open density, is too ray-asymptotic to be globally dense in the supremum norm.

The theorem is stated for real Frechet spaces, matching the source’s separating-property theorem and examples. Since scalar-valued global density already fails, the result also blocks any vector-valued global theorem that would imply the scalar case for this same activation class.

Human Review Recommendation

Reviewers should check that the network class is read exactly as in equation (2) and the subsequent definition of $\mathfrak{N}(\sigma)$ in the source paper. The central proof step is the finite ray limit of each neuron:

$$\ell(\sigma(A(tv) + b)) = \ell(z)\beta(t\psi(Av) + \psi(b)).$$

Once this is accepted, the oscillating test function $f(x) = \sin(\varphi(x)^2)$ gives a direct obstruction to supremum-norm density.

References

[1] F. E. Benth, N. Detering, and L. Galimberti, *Neural Networks in Fréchet spaces*, arXiv:2109.13512v4, 2022. `https://arxiv.org/abs/2109.13512`

[2] G. Cybenko, *Approximation by superpositions of a sigmoidal function*, Mathematics of Control, Signals and Systems **2** (1989), 303–314.